Performing the Integration: t-Channel Compton Amplitude from Feynman Rules

From the Full Integral with δ-Functions to Griffiths Eq. 7.112

PHYS 7502 — Particles and Nuclei 2

1 What This Calculation Does

The Question

Starting from the QED Feynman rules (vertices, propagators, δ-functions), write down the *full* integral expression for the t-channel Compton diagram, and perform the integration over the internal momentum q explicitly to arrive at Griffiths Eq. 7.112.

What We Want to Show

How the integral

$$\int \frac{d^4q}{(2\pi)^4} \Big[\text{spinors} \times \text{vertices} \times \text{propagator} \times \text{polarizations}\Big] \times \delta^4(\text{vertex A}) \times \delta^4(\text{vertex B})$$

reduces to the compact expression $\mathcal{M}_t$ with q fixed to $p_1 - p_2$ and the remaining δ-function enforcing overall 4-momentum conservation.

2 Setup: The QED Feynman Rules We Need

Relevant Feynman Rules (Griffiths Section 7.6)

Rule 1 — External fermion lines:

- Incoming electron: $u(p)$
- Outgoing electron: $\bar{u}(p)$

Rule 2 — External photon lines:

- Incoming photon: $\varepsilon_\mu(p)$
- Outgoing photon: $\varepsilon^*_\mu(p)$

Rule 3 — Vertex factor:
Each QED vertex (two fermion lines, one photon line) contributes:

$$ig_e\gamma^\mu$$

where g_e is the QED coupling ($g_e^2 = 4\pi\alpha = e^2$). The Lorentz index μ contracts with the photon polarization vector or propagator.

Rule 4 — Internal fermion propagator:
A virtual fermion with 4-momentum q contributes:

$$\frac{i(\not{q} + m)}{q^2 - m^2}$$

Rule 5 — Momentum conservation at each vertex:
Each vertex contributes a factor:

$$(2\pi)^4 \, \delta^{(4)}\Big(\sum p_{\text{in}} - \sum p_{\text{out}}\Big)$$

Rule 6 — Integration over internal momenta:
For each internal line, integrate:

$$\int \frac{d^4q}{(2\pi)^4}$$

Rule 7 — Reading order:
Work *backward* along each fermion line (against the arrow), writing factors left to right: $\bar{u} \cdots [\text{vertex}] \cdots [\text{propagator}] \cdots [\text{vertex}] \cdots u$.

3 The t-Channel Diagram: Identifying the Pieces

The t-Channel Diagram (Griffiths Fig. 7.7)

Griffiths labeling: $1 = e^-_{\text{in}}(p_1)$, $2 = \gamma_{\text{out}}(p_2)$, $3 = \gamma_{\text{in}}(p_3)$, $4 = e^-_{\text{out}}(p_4)$.

Physical process in the t-channel: The incoming electron *first emits* the outgoing photon $\gamma(p_2)$, propagates as a virtual electron with momentum q, then *absorbs* the incoming photon $\gamma(p_3)$, and exits as the outgoing electron.

Two vertices:

Vertex A (earlier in time): Incoming $e^-(p_1)$ emits outgoing $\gamma(p_2)$ and becomes virtual $e^-(q)$.

$$\text{In: } e^-(p_1) \qquad \longrightarrow \qquad \text{Out: } e^-_{\text{virt}}(q),\ \gamma(p_2)$$

Momentum conservation: $p_1 = q + p_2$, i.e., $q = p_1 - p_2$.

Vertex B (later in time): Virtual $e^-(q)$ absorbs incoming $\gamma(p_3)$ and becomes outgoing $e^-(p_4)$.

$$\text{In: } e^-_{\text{virt}}(q),\ \gamma(p_3) \qquad \longrightarrow \qquad \text{Out: } e^-(p_4)$$

Momentum conservation: $q + p_3 = p_4$.

Reading Order: Backward Along the Fermion Line

Following Rule 7, we read from $\bar{u}(p_4)$ backward to $u(p_1)$. This means we encounter the vertices in *reverse* time order:

$$\underbrace{\bar{u}(p_4)}_{\text{outgoing } e^-}\ \underbrace{(ig_e\gamma^\mu)}_{\text{Vertex B}}\ \underbrace{\frac{i(\not{q}+m)}{q^2-m^2}}_{\text{propagator}}\ \underbrace{(ig_e\gamma^\nu)}_{\text{Vertex A}}\ \underbrace{u(p_1)}_{\text{incoming } e^-}$$

The index μ at Vertex B contracts with the incoming photon: $\varepsilon_{3\mu}$.
The index ν at Vertex A contracts with the outgoing photon: $\varepsilon^*_{2\nu}$.

4 Writing Down the Full Integral

Assembling all the pieces:

Full Integral Expression for the t-Channel

$$\begin{aligned} &i\mathcal{M}_t \times (2\pi)^4 \delta^{(4)}(p_1+p_3-p_2-p_4) \\ &= \int \frac{d^4q}{(2\pi)^4} \left[\bar{u}(p_4)\ \underbrace{(ig_e\gamma^\mu)}_{\text{Vertex B}}\ \underbrace{\frac{i(\not{q}+m)}{q^2-m^2}}_{\text{propagator}}\ \underbrace{(ig_e\gamma^\nu)}_{\text{Vertex A}}\ u(p_1) \right] \\ &\quad \times\ \varepsilon_{3\mu}(p_3)\ \times\ \varepsilon^*_{2\nu}(p_2) \\ &\quad \times\ \underbrace{(2\pi)^4\,\delta^{(4)}(p_1-q-p_2)}_{\text{Vertex A conservation}} \\ &\quad \times\ \underbrace{(2\pi)^4\,\delta^{(4)}(q+p_3-p_4)}_{\text{Vertex B conservation}} \end{aligned} \tag{1}$$

Counting the Factors of (2π)

- Each vertex contributes a $(2\pi)^4\delta^{(4)}$: that is $2 \times (2\pi)^4$.
- Each internal line contributes an integral $\int d^4q/(2\pi)^4$: that is $1 \times 1/(2\pi)^4$.
- Net: $(2\pi)^4 \times (2\pi)^4 \times \frac{1}{(2\pi)^4} = (2\pi)^4$.

This leftover $(2\pi)^4$ accompanies the overall momentum-conserving δ-function on the left-hand side. It will cancel when we extract $\mathcal{M}_t$ (which is defined without this factor).

5 Performing the q Integration

Step 1: Use the First δ-Function to Fix q

The Vertex A δ-function is:

$$\delta^{(4)}(p_1-q-p_2)$$

This sets $q = p_1 - p_2$. The standard identity is:

$$\int \frac{d^4q}{(2\pi)^4}\, f(q) \;\cdot\; (2\pi)^4 \delta^{(4)}(p_1 - p_2 - q) = f(p_1 - p_2)$$

Applying this to (1), the integral over q is performed, and everywhere q appeared, we substitute $q = p_1 - p_2$:

$$= \left[\bar{u}(p_4)\,(ig_e\gamma^\mu)\,\frac{i(\not{p}_1 - \not{p}_2 + m)}{(p_1 - p_2)^2 - m^2}\,(ig_e\gamma^\nu)\,u(p_1)\right]\varepsilon_{3\mu}\,\varepsilon^*_{2\nu}$$
$$\times\; (2\pi)^4\,\delta^{(4)}\big(\underbrace{(p_1 - p_2)}_{=q} + p_3 - p_4\big) \tag{2}$$

What Just Happened

The integral over d^4q was trivial—the δ-function at Vertex A *pinned* the internal momentum to $q = p_1 - p_2$. This is the defining feature of tree-level diagrams: there are no free loop momenta. Every internal momentum is determined by the external momenta through the vertex δ-functions.

Step 2: The Remaining δ-Function is Overall Conservation

The remaining δ-function from Vertex B becomes:

$$\delta^{(4)}(p_1 - p_2 + p_3 - p_4)$$

This is simply the **overall 4-momentum conservation**:

$$p_1 + p_3 = p_2 + p_4$$

which was guaranteed from the start. We factor this out along with its $(2\pi)^4$ to define $\mathcal{M}_t$.

Step 3: Simplify the Coupling Constants

The two vertex factors give:

$$(ig_e)(ig_e) = i^2 g_e^2 = -g_e^2$$

Combined with the i from the propagator numerator:

$$(-g_e^2) \times i = -ig_e^2$$

And this multiplies the $i\mathcal{M}_t$ on the left-hand side, so $\mathcal{M}_t$ picks up:

$$\frac{(ig_e)^2 \cdot i}{1} = \frac{-g_e^2 \cdot i}{1} \quad \Rightarrow \quad i\mathcal{M}_t \text{ gives } \mathcal{M}_t = \frac{-g_e^2}{(p_1 - p_2)^2 - m^2}(\ldots)$$

More explicitly: from (2), stripping off the overall $(2\pi)^4\delta^{(4)}$:

$$i\mathcal{M}_t = (ig_e)^2\,\varepsilon_{3\mu}\,\varepsilon^*_{2\nu}\,\bar{u}(p_4)\,\gamma^\mu\,\frac{i(\not{p}_1 - \not{p}_2 + m)}{(p_1 - p_2)^2 - m^2}\,\gamma^\nu\,u(p_1)$$

$$= -g_e^2\,\varepsilon_{3\mu}\,\varepsilon^*_{2\nu}\,\frac{\bar{u}(p_4)\,\gamma^\mu\,(\not{p}_1 - \not{p}_2 + m)\,\gamma^\nu\,u(p_1)}{(p_1 - p_2)^2 - m^2} \;\times\; i \tag{3}$$

Dividing both sides by i (and writing $g_e^2 = e^2$, $(p_1 - p_2)^2 = t$):

6 Final Result: Griffiths Eq. 7.112

t-Channel Compton Amplitude (Griffiths Eq. 7.112)

$$\boxed{\mathcal{M}_t = \frac{-e^2}{t - m^2}\, \bar{u}(p_4)\, \not{\varepsilon}_3\, (\not{p}_1 - \not{p}_2 + m)\, \not{\varepsilon}_2\, u(p_1)} \tag{4}$$

where:

- $\not{\varepsilon}_3 \equiv \varepsilon_{3\mu}\gamma^\mu$ is the incoming photon polarization (contracted at Vertex B),
- $\not{\varepsilon}_2 \equiv \varepsilon^*_{2\nu}\gamma^\nu$ is the outgoing photon polarization (contracted at Vertex A),
- $\not{p}_1 - \not{p}_2 + m$ is the propagator numerator with $q = p_1 - p_2$,
- $t - m^2 = (p_1 - p_2)^2 - m^2$ is the propagator denominator.

7 Comparison with the s-Channel

For completeness, the same procedure applied to the s-channel diagram (Griffiths Fig. 7.6, Eq. 7.111) gives:

	s-channel (Eq. 7.111)	**t-channel (Eq. 7.112)**
Vertex order	absorb $\gamma(p_3)$, then emit $\gamma(p_2)$	emit $\gamma(p_2)$, then absorb $\gamma(p_3)$
δ-function fixes q to	$q = p_1 + p_3$	$q = p_1 - p_2$
Propagator momentum	$\not{p}_1 + \not{p}_3 + m$	$\not{p}_1 - \not{p}_2 + m$
Denominator	$s - m^2 = (p_1 + p_3)^2 - m^2$	$t - m^2 = (p_1 - p_2)^2 - m^2$
Polarization order	$\not{\varepsilon}_2 \cdots \not{\varepsilon}_3$	$\not{\varepsilon}_3 \cdots \not{\varepsilon}_2$
$\mathcal{M}$	$\frac{-e^2}{s-m^2}\bar{u}(4)\not{\varepsilon}_2(\not{p}_1 + \not{p}_3 + m)\not{\varepsilon}_3\, u(1)$	$\frac{-e^2}{t-m^2}\bar{u}(4)\not{\varepsilon}_3(\not{p}_1 - \not{p}_2 + m)\not{\varepsilon}_2\, u(1)$

Why the Polarizations Swap

In the s-channel, reading backward along the fermion line, you hit the outgoing photon vertex *first* (giving $\not{\varepsilon}_2$), then the propagator, then the incoming photon vertex (giving $\not{\varepsilon}_3$). In the t-channel, the time ordering of the photon interactions is reversed: you hit the incoming photon *first* ($\not{\varepsilon}_3$), then the propagator, then the outgoing photon ($\not{\varepsilon}_2$).
The swap $\not{\varepsilon}_2 \cdots \not{\varepsilon}_3 \;\leftrightarrow\; \not{\varepsilon}_3 \cdots \not{\varepsilon}_2$ is the signature difference between the two channels.

8 Summary: The Integration in Three Lines

The Whole Procedure

1. Write the full integral using Feynman rules (vertices $\times$ propagator $\times$ δ-functions $\times$ $\int d^4q/(2\pi)^4$):

$$\int \frac{d^4q}{(2\pi)^4} \left[\bar{u}(4)(ig_e\gamma^\mu)\frac{i(\not{q}+m)}{q^2-m^2}(ig_e\gamma^\nu)u(1)\right] \varepsilon_{3\mu}\,\varepsilon^*_{2\nu} \times (2\pi)^4\delta^{(4)}(p_1-q-p_2) \times (2\pi)^4\delta^{(4)}(q+p_3-p_4)$$

2. Integrate over q using $\delta^{(4)}(p_1-q-p_2)$, which sets $\boldsymbol{q=p_1-p_2}$:

$$\left[\bar{u}(4)(ig_e\gamma^\mu)\frac{i(\not{p}_1-\not{p}_2+m)}{(p_1-p_2)^2-m^2}(ig_e\gamma^\nu)u(1)\right] \varepsilon_{3\mu}\,\varepsilon^*_{2\nu} \times (2\pi)^4\delta^{(4)}(p_1+p_3-p_2-p_4)$$

3. Factor out the overall $(2\pi)^4\delta^{(4)}$ and simplify couplings:

$$\mathcal{M}_t = \frac{-e^2}{t-m^2}\,\bar{u}(p_4)\,\not{\varepsilon}_3\,(\not{p}_1-\not{p}_2+m)\,\not{\varepsilon}_2\,u(p_1) \qquad \checkmark$$